Evaluation runs: **vrr** (vrr)

generated 2026-11-08 from brain-data e1e40f443 2026-11-08 13:42:54 +0100

Every campaign run on this benchmark, one row per instance (variant, specification, communication model, size) and engine; the newest run of an instance wins. Expected verdicts (the rule file): no entry in the rule file (the README states the expectation). Blue rows are at the current pins (TraceForge 367cdc6, whose predecessor 9a0629e is equivalent for every engine but the join; merge join 367cdc6; JMC 3c43b2d2): a rerun gives the same result. Red rows are superseded (an older pin, or the in-memory join that the merge join replaces) and their rerun is pending. A planned instance without a measurement is not a row: it is listed under "Not run". G_{ref}, G_{cand} = the graphs of the reference and of the implementation (the stateless engine's own count of specification graphs explored, repeats included, is G_{ref}^{expl}); G_x = the implementation graphs no specification graph covers (violating pairs); T = wall seconds; M = peak resident MiB; tier = the run's time and memory limits; * marks a verdict other than the expected one; 🕓 = the time tier was reached; oom = the memory limit was reached.

stateless

stateless: pair **single**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	sweeps	G_{ref}^{expl}	run
3,2	2	868	conforms	0.1	2	4 h, 250 GB	2	3	brain01, 2026-11-03
3,3	6	1194756	conforms	20.4	21	4 h, 250 GB	7	21	brain01, 2026-11-03

stateful

stateful: pair **single**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	run
3,2	2	868	conforms	0.1	2	4 h, 250 GB	brain01, 2026-11-03
3,3	6	1194756	conforms	20.4	22	4 h, 250 GB	brain01, 2026-11-03
3,4	24	6039877584	conforms	46587.0	15737	48 h, 250 GB	brain03, 2026-11-03
4,3	6	8284096548	conforms	116784.0	21740	48 h, 80 GB	pinky15, 2026-11-05

offline

offline: pair **single**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
3,2	2	868	conforms	0.1	8	1 h, 250 GB	0.1	0.1	0.0	merge	brain01, 2026-11-03
3,3	6	1194756	conforms	21.3	25	1 h, 250 GB	0.1	21.3	0.0	merge	brain01, 2026-11-03
3,4	24	6039877584	conforms	50027.0	15670	48 h, 250 GB	0.0	50027.0	0.0	merge	brain03, 2026-11-07
4,3	6	8284096548	conforms	63371.0	21748	48 h, 250 GB	0.0	63371.0	0.0	merge	brain03, 2026-11-04

cross offline, Java implementation against the Rust specification

cross offline, Java implementation against the Rust specification: pair **single**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
3,2	2	868	conforms	11.9	2209	4 h, 250 GB	0.1	11.9	0.0	in-memory	brain01, 2026-11-03

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
3,3	6	1194756	conforms	4059.0	26800	4 h, 250 GB	0.1	4059.0	0.0	in-memory	brain01, 2026-11-03

Not run (cross offline, Java implementation against the Rust specification). skipped: a smaller size had reached its limit: 3 4 bag, 4 3 bag.

cross online, Java implementation (JMC) against the Rust specification export

cross online, Java implementation (JMC) against the Rust specification export: pair **single**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
3,2	2	868	conforms	12.0	2162	4 h, 250 GB		brain01, 2026-11-03
3,3	6	1194756	conforms	2861.9	22854	4 h, 250 GB		brain01, 2026-11-03

context-free twins offline, Rust implementation against the context-free Java port

context-free twins offline, Rust implementation against the context-free Java port: pair **single**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
3,2	868	868	conforms	6.4	238	4 h, 250 GB	6.3	0.1	0.1	merge	brain01, 2026-11-03
3,3	1194756	1194756	conforms	1426.5	15506	4 h, 250 GB	1426.1	21.3	0.4	merge	brain01, 2026-11-03

Not run (context-free twins offline, Rust implementation against the context-free Java port). skipped: a smaller size had reached its limit: 3 4 bag, 4 3 bag.

context-free twins offline, Java port against the Rust implementation

context-free twins offline, Java port against the Rust implementation: pair **single**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
3,2	868	868	conforms	6.4	238	4 h, 250 GB	0.1	6.3	0.1	merge	brain01, 2026-11-03
3,3	1194756	1194756	conforms	1426.1	15506	4 h, 250 GB	21.3	1426.1	0.0	merge	brain01, 2026-11-03

Not run (context-free twins offline, Java port against the Rust implementation). skipped: a smaller size had reached its limit: 3 4 bag, 4 3 bag.

context-free twins online, Java port (JMC) against the Rust implementation's export

context-free twins online, Java port (JMC) against the Rust implementation's export: pair **single**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
3,2	868	868	conforms	10.5	2099	4 h, 250 GB		brain01, 2026-11-03
3,3	1194756	1194756	conforms	2998.5	23725	4 h, 250 GB		brain01, 2026-11-03

context-free twins online, Rust implementation (TraceForge) against the Java port's export

context-free twins online, Rust implementation (TraceForge) against the Java port's export: pair **single**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
3,2	868	868	conforms	11.9	2205	4 h, 250 GB		brain01, 2026-11-03
3,3	1194756	1194756	conforms	4084.2	26796	4 h, 250 GB		brain01, 2026-11-03

Not run (context-free twins online, Rust implementation (TraceForge) against the Java port's export).
skipped: a smaller size had reached its limit: 4 3 bag.

planned but never scheduled (grid points beyond the ladder): 3 4 bag.